LING324 Cognitive Linguistics

Professor Christopher Hart

Lecture 3: Image Schemas

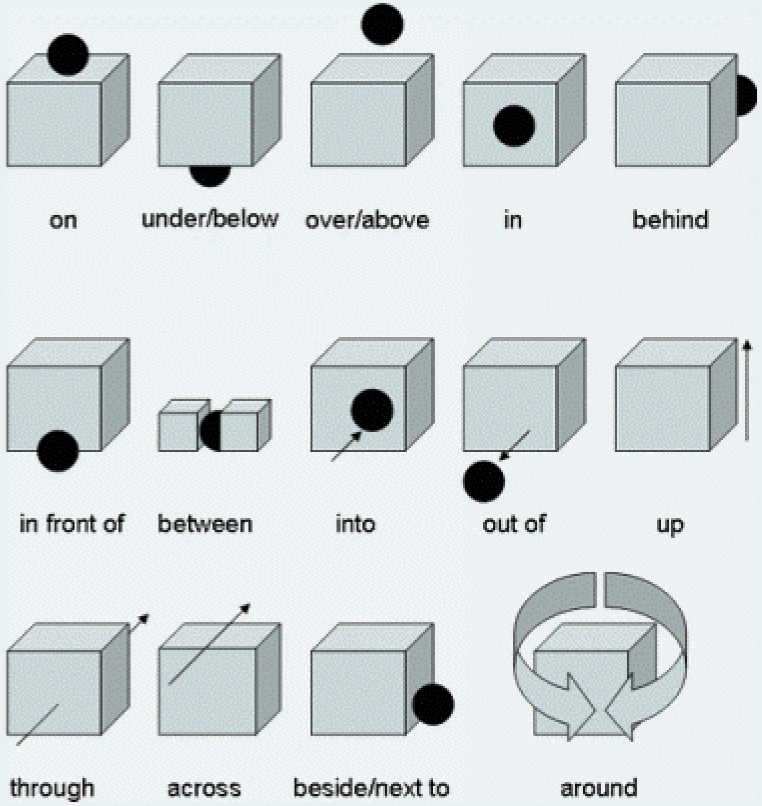
Lancaster University

Overview

- Embodiment
- Image schemas
- Words as radial categories: prepositions
  - *Over*
  - Other prepositions
- Principled polysemy
- Functional geometry
  - *In* vs. *on*
- Cross linguistic

Introduction

- **Frames** = **idealised cognitive model** derived from socio-cultural experience
- **Image schemas** = idealised cognitive model derived from perceptual and sensory-motor experience
- Represent distinct senses of **prepositions** organised in radial networks
- Prepositions proven fruitful site for investigating link between language, cognition and embodied spatial experience



Embodied Cognition

- Rejection of Cartesian **dualism** - “Human cognition is fundamentally shaped by embodied experience” (Gibbs 2005: 3)

> The properties of an organism’s body limit or constrain the concepts an organism can acquire. That is, the concepts on which an organism relies to understand its surrounding world depend on the kind of body that it has. (Shapiro 2011: 4)

- Conceptual structures that provide meaning to language arise from and reflect embodied experience
- Language grounded in perception and action

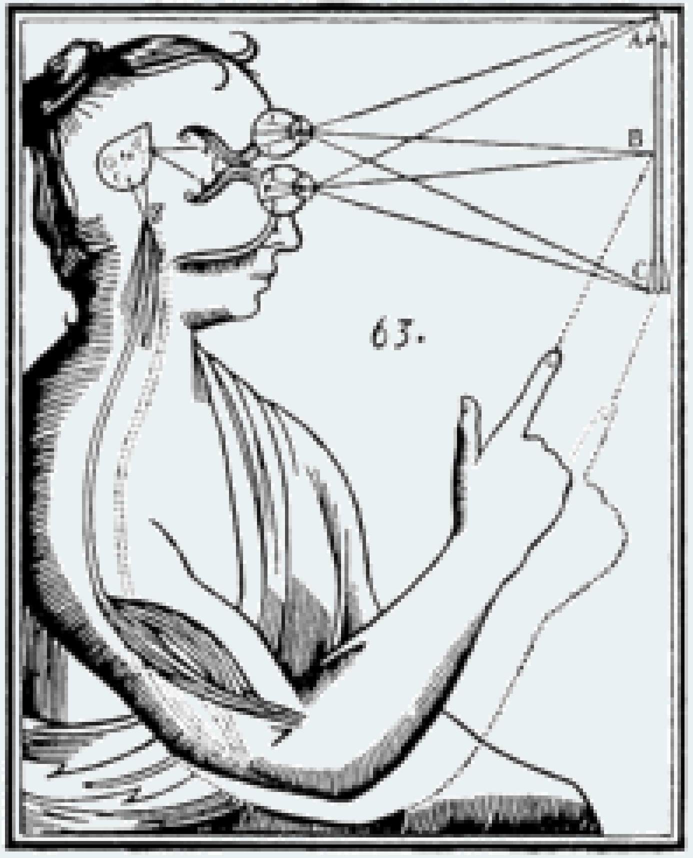
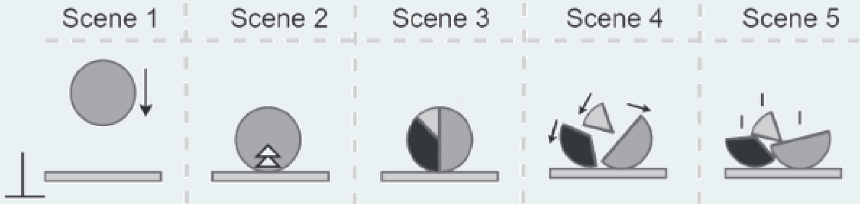
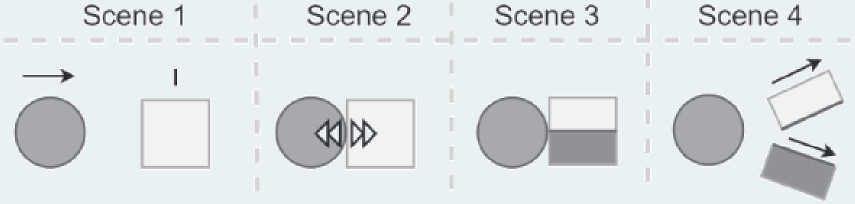


Image Schemas

- Conceptual structures with embodied basis include **image schemas** (Mandler 2004)
- Conceptual structures representing recurrent patterns in perceptual and motor experience
- Derived by process of abstraction from interactions with and observations of physical world
- Not images *per se* but distillations over experience
- Represent common ‘skeletal’ structures which similarly conceived objects, events, situations and relations can be ‘boiled down’ to
- Emerge pre-linguistically as chief building blocks of the mind



(a) A falling object splits into parts on impact.



(b) An object crashes into another object that splits.

(Hedblom, Neuhaus & Mossakowski 2024: 23)

Image Schemas

- Organise experience in basic **domains** like SPACE, MOTION, FORCE and ACTION
- Co-opted by language to provide semantic structure in symbolic units (Johnson 1987; Lakoff 1987)
- Conceptual basis of **prepositions** but also grammatical constructions (Talmy 2000; Langacker 2008)
- Part of frame-based knowledge for nouns and verbs
- Source in metaphorical projections (Lakoff & Johnson 2003)

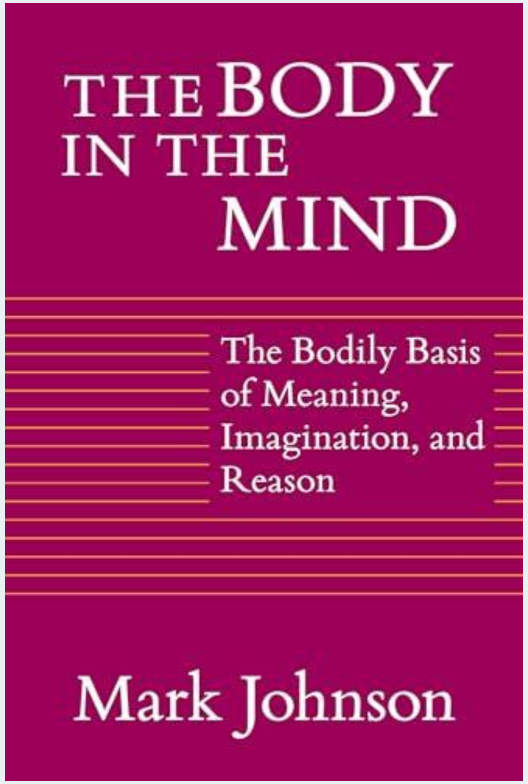
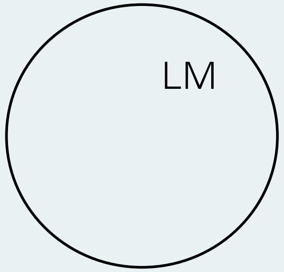
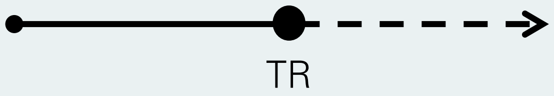


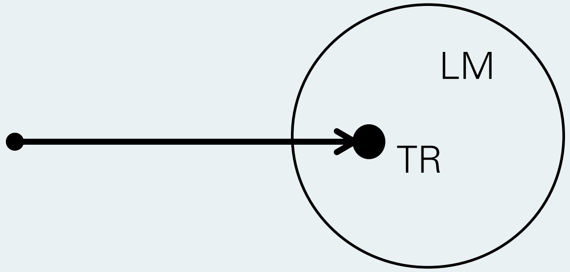
Image Schemas



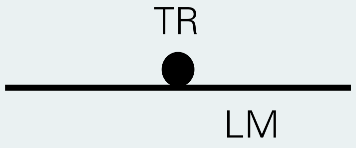
CONTAINER



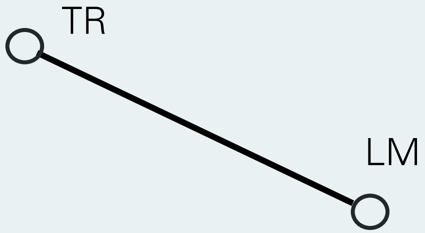
SOURCE-PATH-GOAL



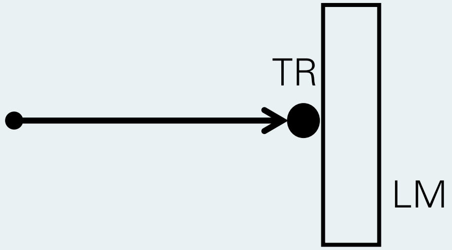
MOTION INTO A CONTAINER



SUPPORT



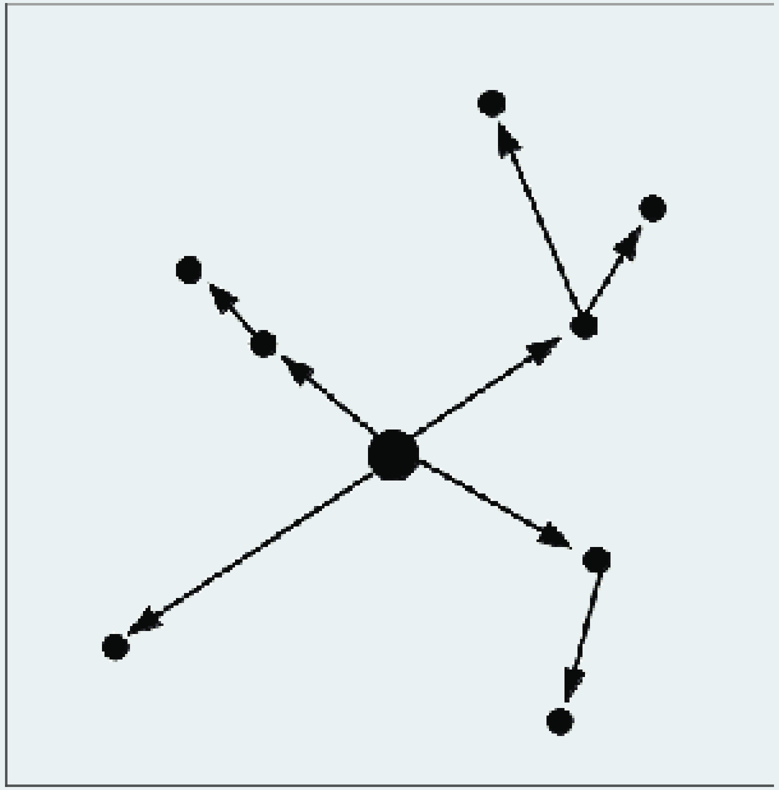
LINK



BLOCKAGE

Words as Radial Categories

- Prepositions highly **polysemous**
  - Polysemy vs. monosemy
- Senses organised in **radial networks** relative to **proto-sense**
- Extension to derived senses **motivated**
  - Spatial and metaphorical extension
  - Involves image schema **transformation**
- Closely related senses organised in **clusters**
- Range of extensions available so network conventional for given language

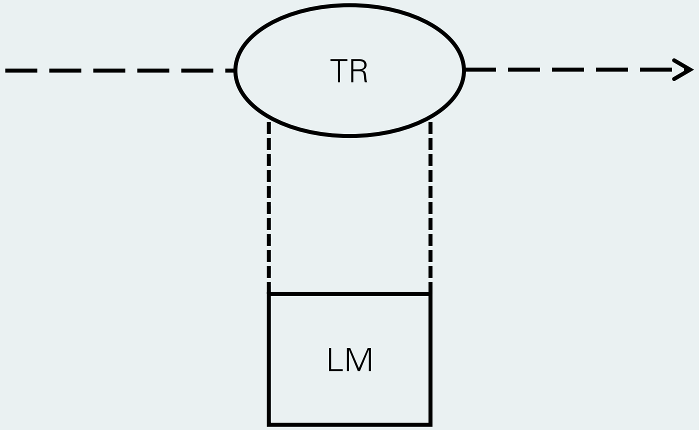


Over

- Grammatically complex – preposition, particle, adverb, prefix – but senses semantically linked (Brugman and Lakoff 1988; Lakoff 1987)

    - (1) The plane flew over

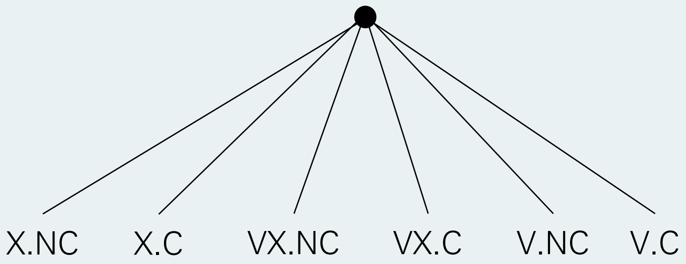
- **Protosense** combines elements of ABOVE and ACROSS senses
- Neutral on issue of contact (Lakoff 1987: 419)



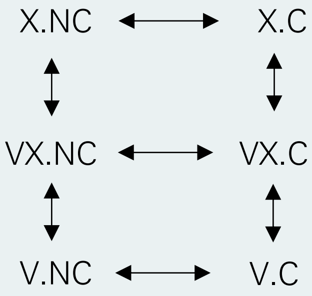
(Lakoff 1987: 419)

Over

- **Similarity links**: instances arrived at by adding information
    - LM as point
    - LM is extended (X)
    - LM is vertical (V)
    - LM extended + vertical
    - Contact (C) vs. no contact (NC)
- Minimal vs. full specification



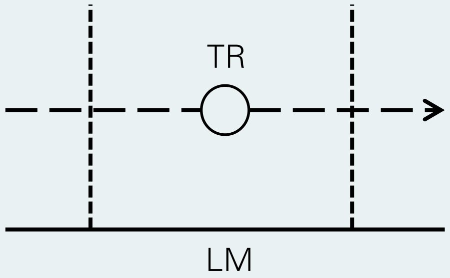
Minimal specification



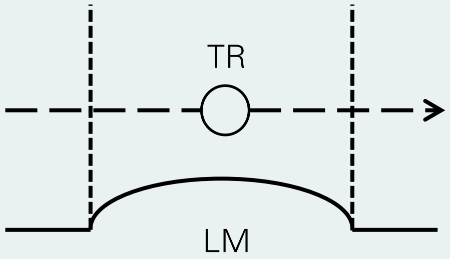
Full specification

(Lakoff 1987: 423)

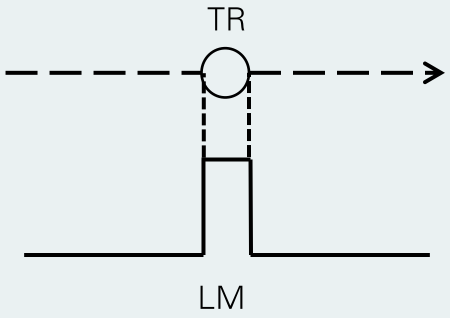
Over



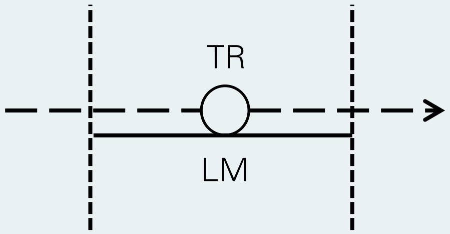
(2) The bird flew over the yard (X.NC)



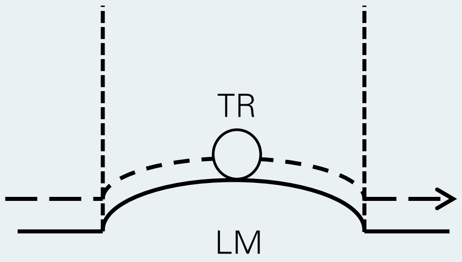
(3) The plane flew over the hill (VX.NC)



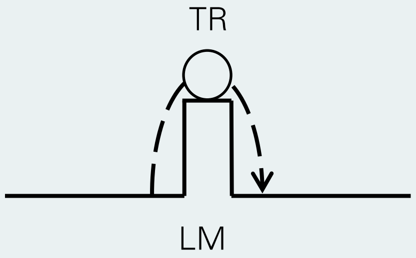
(4) The bird flew over the wall (V.NC)



(5) Wilbur drove over the bridge (X.C)



(6) Ivy walked over the hill (VX.C)

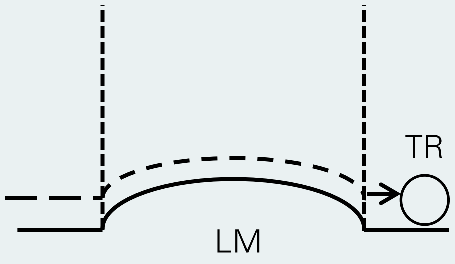


(7) Milo climbed over the wall (V.C)

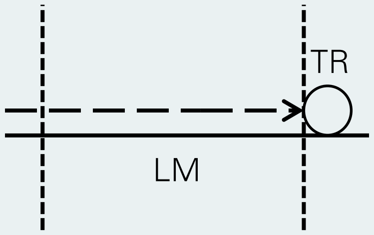
(Lakoff 1987: 421-422)

Over

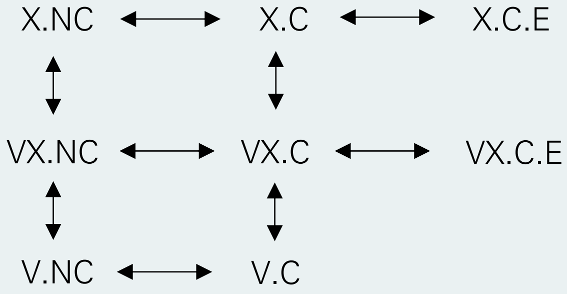
- **Transformational links:** instances arrived at by transformation
  - E.g. **endpoint focus**



(8) Ivy lives over the hill (VX.C.E)



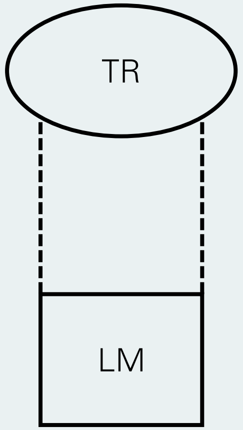
(9) The restaurant is over the bridge (X.C.E)



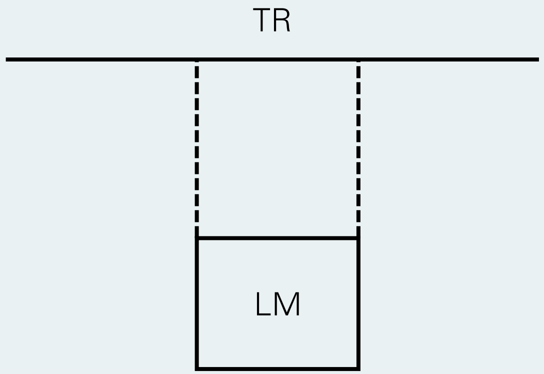
(Lakoff 1987: 423-433)

Over

- ABOVE senses:



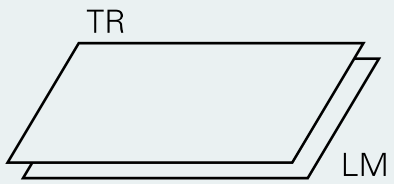
(10) The painting hangs over the fireplace



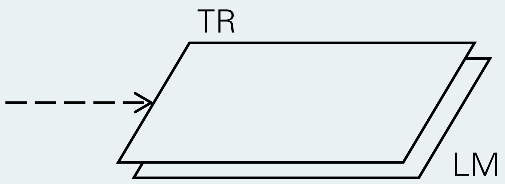
(11) The power line stretches over the yard

Over

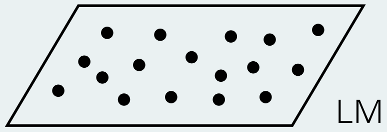
- Other senses: COVERAGE (12-15) and REFLEXIVE (16)



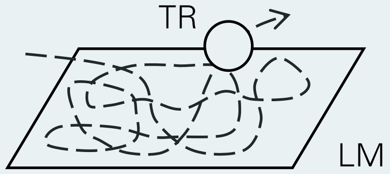
(12) The board is over the hole



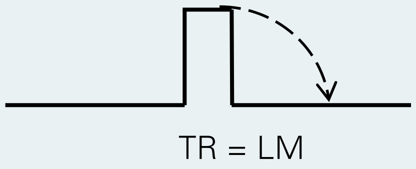
(13) The city clouded over



(14) The guards were posted all over the hill



(15) I walked all over the hill

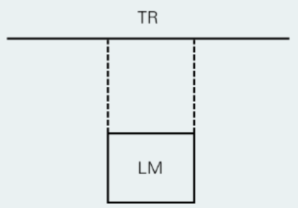


(16) The fence fell over

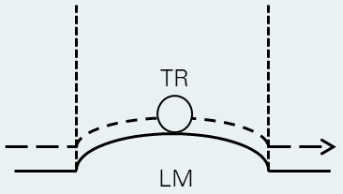
(Lakoff 1987: 426-433)

Over

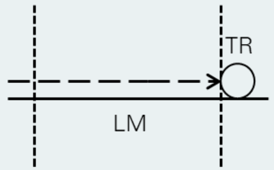
- Metaphorical extensions:
  - (17) She holds a strange power over me



  - (18) Ron is still not over his divorce

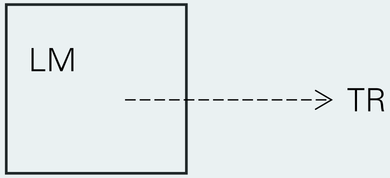


  - (19) The play is over

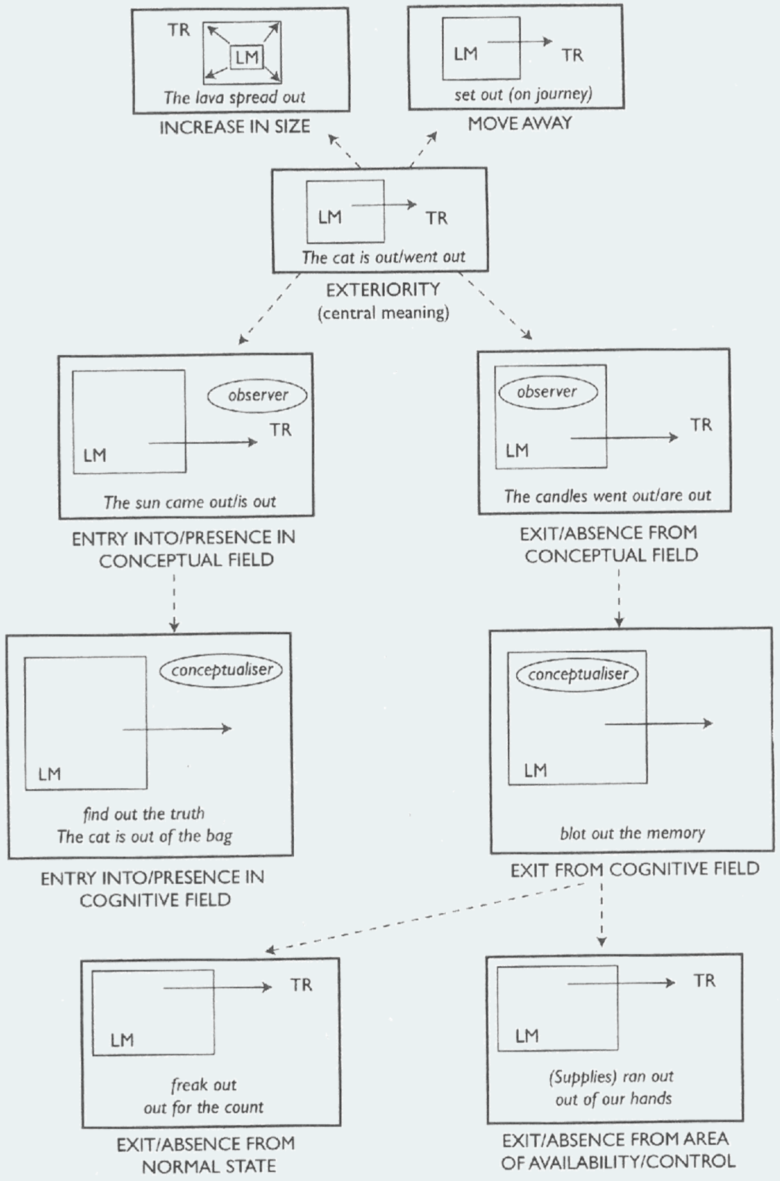


Other Prepositions: *Out*

- Same kind of analysis can be given of other prepositions (e.g. Tyler & Evans 2003)
    - E.g. *out* (Lee 2001)



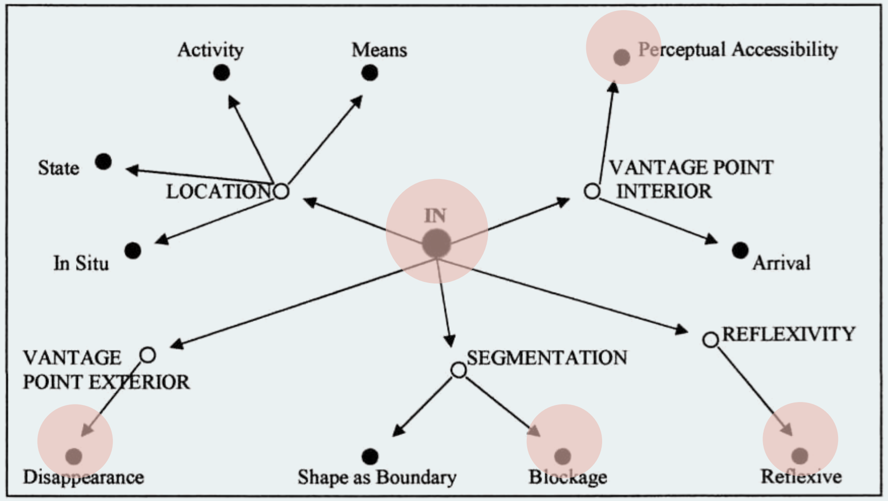
(20) The cat is out/went out



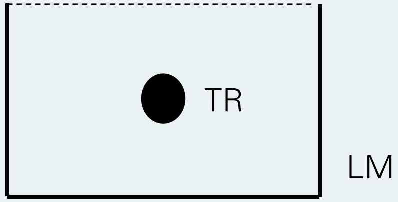
(Lee 2001: 35)

Other Prepositions: *In*

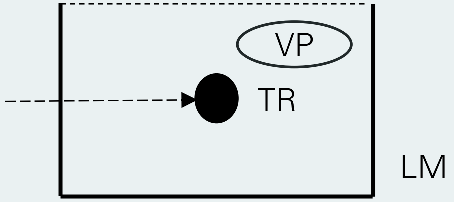
- LMs construed as CONTAINER motivated by commonalities with protoscene (Evans & Tyler 2004; Tyler & Evans 2003)



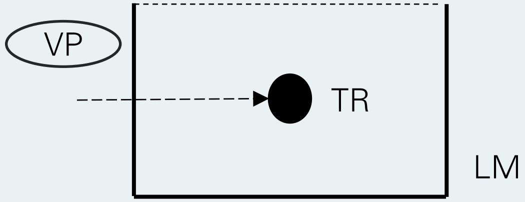
(Evans & Tyler 2004)



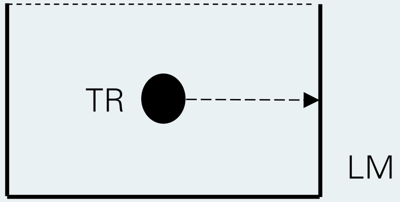
(21) The toys are in the box



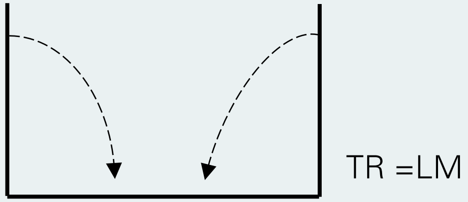
(22) He's now in sight



(23) The sun went in



(24) The car is blocked in



(25) The house caved in

Principled Polysemy

- Full specification vs. principled polysemy
  - All senses stored conceptually or some derived contextually?
  - E.g. contact need not yield distinct senses as inferable from encyclopaedic knowledge

    (26) a. Ivy jumped over the wall b. Ivy climbed over the wall

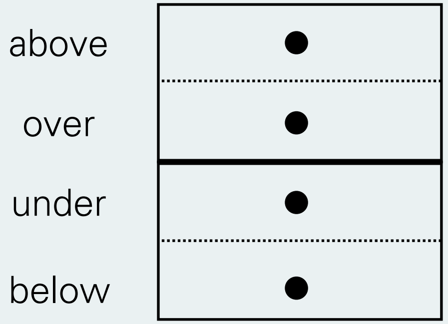
  - What counts as protosense?

    (27) a. The plane flew over b. The painting hangs over the fireplace

  - Senses established on analyst intuition – no principled criteria

Principled Polysemy

- Principled polysemy approach provides criteria for establishing proto-sense and distinct extended senses (Tyler & Evans 2001, 2003)
- Protosense:
    - Earliest attested meaning
    - Predominant within the network
    - Contrast with other prepositions
    - Best predictor of other senses



Principled Polysemy

- Distinct senses:

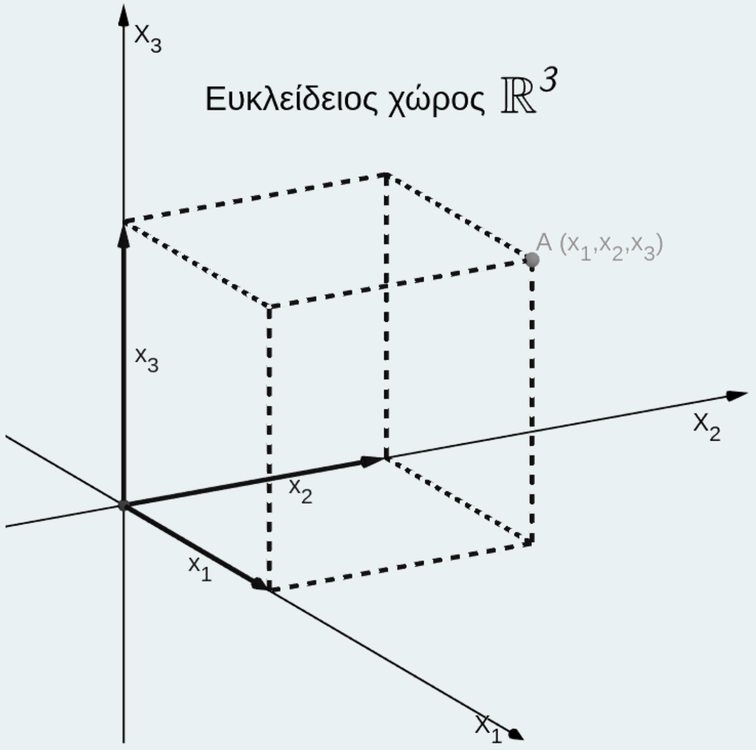
  I. Must contain additional meaning not apparent in any other senses associated with a particular form, that is, must involve non-spatial meaning or a different configuration between the TR and LM than found in the proto-scene

  II. Must be instances of the sense that are context independent, that is, in which the distinct sense could not be inferred from another sense and the context in which it occurs

(Tyler & Evans 2003: 42-43)

Functional Geometry

- The meaning of a preposition includes functional as well as geometric information (Coventry & Garrod 2004; Garrod, Ferrier & Campbell 1999)
- How objects interact with one another as important to language users as relations in Euclidean space
- Meaning of a preposition includes **affordances** of spatial arrangements
- **Functional geometric framework** – "combines both geometric and extra-geometric constraints to establish the situation-specific meaning of spatial expressions" (Coventry & Garrod 2004: 52)



Functional Geometry

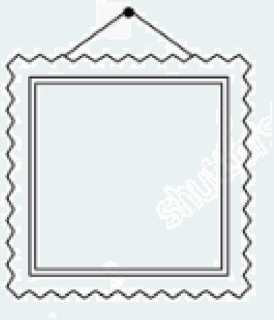
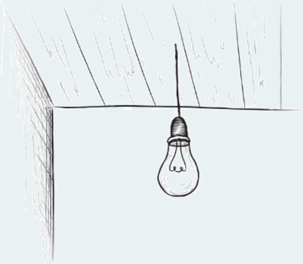
- Relations of enclosure and contact encoded by *in* and *on* afford different degrees of location control

> "Inness" is a hybrid relation that involves both a geometric containment relation and a certain kind of control relation whereby a container constrains the location of its contents.
> (Coventry & Garrod 2004: 53)

- Explains non-prototypical uses of *in* and *on*, e.g. where TR only loosely or not fully enclosed by LM
- Motivated by functional rather than geometric relations



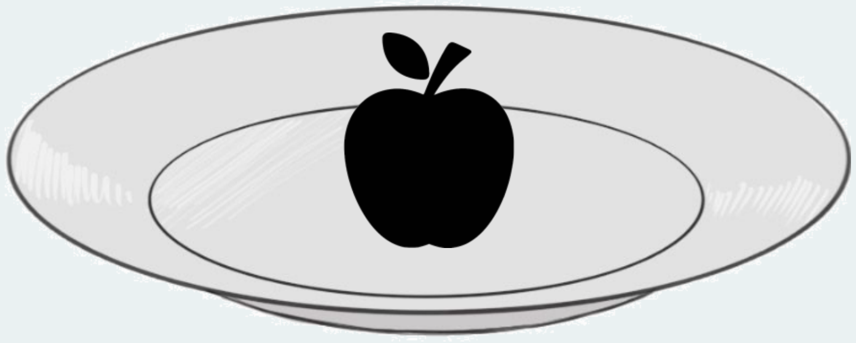


Functional Geometry

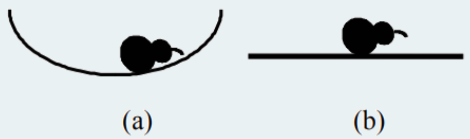
- Two sources of extra-geometric information
  - Force-dynamic routines
  - Object knowledge

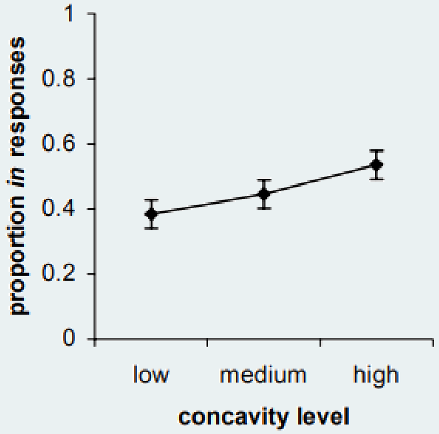
What objects are fundamentally influences how one talks about *where* they are located. (Coventry & Garrod 2004: 55)



Functional Geometry

- Psycholinguistic evidence for extra-geometric meaning
- Appropriateness of *in* and *on* influenced by geometric relations with different functional consequences
    - Concavity of LM (Feist & Gentner 1998)
        - *In* rated more appropriate for LMs with high concavity levels while *on* rated more appropriate for flatter LMs
    - Orientation of LM (Coventry 1998; Coventry & Prat-Sala 2001)
        - Greater degree of tilt in LM, lower acceptability ratings for *in* and *on*
    - Animacy of TR (Feist & Gentner 1998)
        - Ratings for *in* lower for animate TRs than inanimate TRs

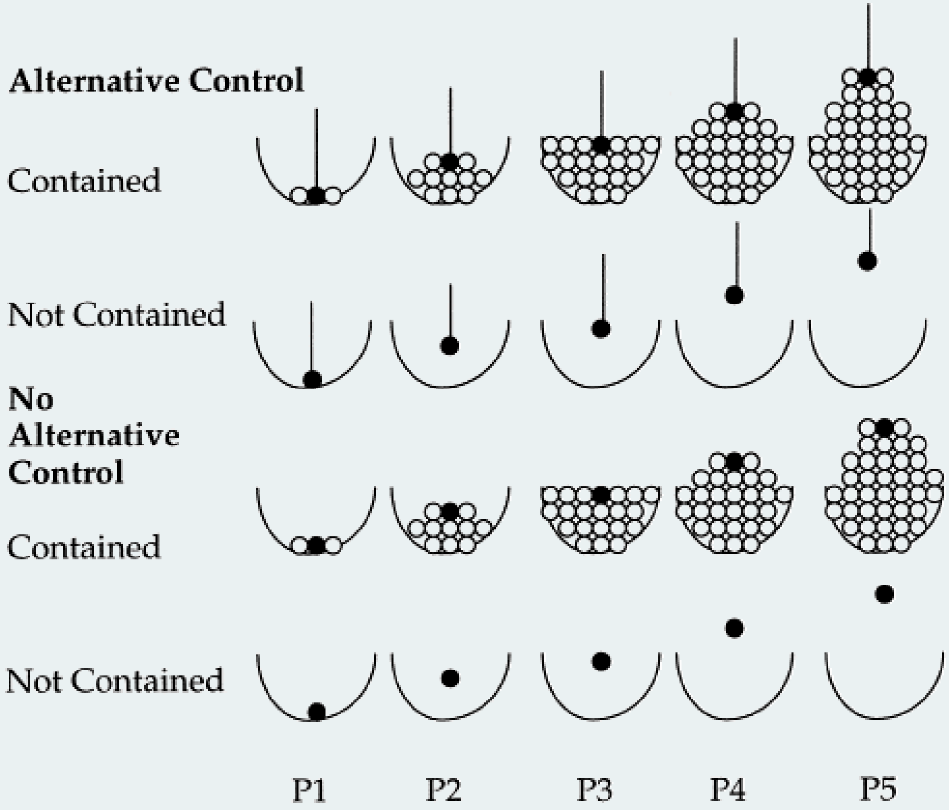




(Feist & Gentner 1998)

Functional Geometry

- Direct evidence for correlation between geometric and extra-geometric factors in language users' understanding of prepositions (Garrod et al. 1999)
- Manipulations made:
    - Geometric relation between ball and bowl itself (P1-5)
    - Degree to which ball surrounded by other balls
    - Whether or not attached to alternative source of control

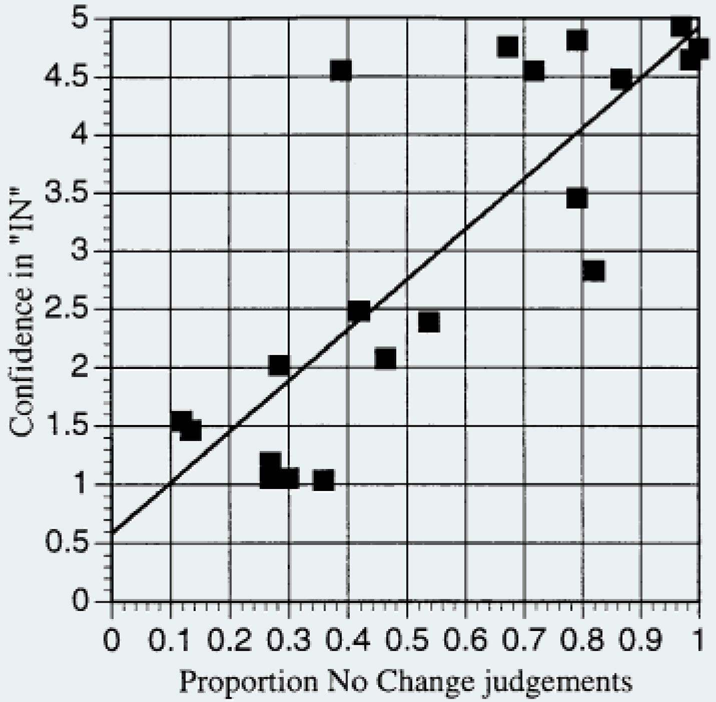


(Garrod et al. 1999: 175)

Functional Geometry

- Subjects indicated:
  - Whether or not black ball would change location if bowl moved (location control)
  - Confidence in use of *in*

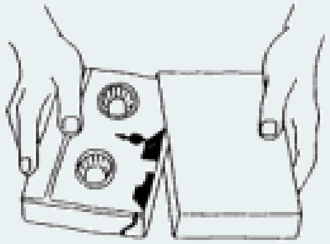
this result clearly implicates location control in the representation of the semantics of *in*
(Garrod et al. 1999: 179)

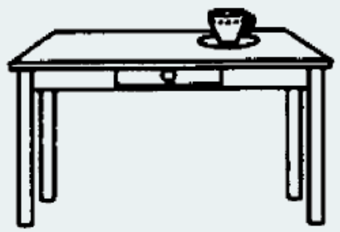


(Garrod et al. 1999: 180)

Cross-Linguistic

- Conceptual organisation of space not reflective of objectively held relations
- Number and extension of categories differs across languages
- Spatial discriminations not made in English made in other languages
  - Korean tight vs. loose fit containment
  - Dutch point vs. surface contact





Thank you

Selected References

Coventry, K. and Garrod, S. (2004). *Saying, Seeing and Acting: The Psychological Semantics of Spatial Prepositions*. Psychology Press.

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